CHAPTER-1

CHAPTER – 1

INTRODUCTION OF THE STUDY

1.1 INTRODUCTION

Field work is the great opportunity for students who want to gain a new experience over the summer at a small-to-medium enterprise. The internship opportunities enable me to use what we have learned and to expend our knowledge and benefits on-the-job experience. There is a great way to explore the career choices that lie ahead of such training. Field work offer students a period of practical experience in the industry relating to their field of study this experience is variable to students as a means of allowing them to experience that can be highly attractive to potential employers on a candidate.

A study on functional areas in the deals with all the functional departments of the company. All the departments play a vital role in the company, each departments perform separate task but they are inter related with each other. In order to make the work efficient the individual should be trained well.

The employees are given training and also motivated in the organization to do their work very efficiently. The industrial training is also helpful to acquire new manipulative skills, technical knowledge, problem solving approach and helps to have a better understanding of the organization.

Most of the students apply for mini project during their summer and winter breaks. In some universities, Field work during the college breaks is compulsory and a part of curriculum. It is common that previous interns would become employees to the organization once they have acquired the necessary skills and experience. Hence technically should not be termed as Field work program.

1.2 OBJECTIVES OF THE STUDY

- To study the functions of Nova Carbon India Pvt. Ltd.
- To know the demographic characteristics of the employees in Nova Carbon India Pvt. Ltd.
- To get out the industrial activities and also about the various functional department in the organizations.
- To boost up the skills and widen horizon of thinking.
- To get industrial exposure the real organization setup.

CHAPTER-2

CHAPTER – 2

2.1 INTRODUCTION

Activated carbon is also called as activated charcoal, is a form of carbon commonly used to filter contaminants from water and air, among many other uses. It is processed to have small, low-volume pores that increase the surface area available for adsorption or chemical reactions. Activation is analogous to making popcorn from dried corn kernels: popcorn is light, fluffy and its kernels have a high surface-area-to-volume ratio. Activated is sometimes replaced by active.

Due to its high degree of microporosity, one gram of activated carbon has a surface area in excess of 3.000 m as determined by gas adsorption. Charcoal, before activation, has a specific surface area in the range of 2.0-5.0 m/g. An activation level sufficient for useful application maybe obtained solely from high surface area. Further chemical treatment often enhances adsorption properties.

Activated carbon is usually derived from waste products such as coconut husks; waste from paper mills has been studied as a sources. These bulk sources are converted into charcoal before being activated. When derived from coal it is referred to as activated coal.

Activated carbon is used in methane and hydrogen storage, air purification, capacitive deionzation, supercapacitive swing adsorption, solvent recovery, decaffeination, gold purification, metal extaction, water purification, medicine, sewage treatment, air filters in respirators, filters in compressed air, teeth whitening, production of hydrogen chloride, edible electronics and many other applications.

INDUSTRIAL

One major industrial application involves use of activated carbon in metal finishing for purification of electroplating solutions. For example, it is the main purification techniques for removing organic impurities from bright nickel plating solution. A variety of organic chemicals are added to plating solution for improving their deposit qualities and for enhancing properties like brightness, smoothness, ductility, etc. Due to passage of direct current and electrolytic reactions of anodic oxidation and cathodic reduction. Organic

additives generate unwanted breakdown products in solution. Their excessive build up can adversely affect plating quality and physical properties of deposited metal. Activated carbon treatment removes such impurities and restores plating performance to the desired level.

MEDICAL

Activated carbon is used to treat poisonings and overdoses following oral ingestion. Tablets or capsules of activated carbon are used in many countries as an over-the-counter drug to treat diarrhea, idigestion and flatulence. However activated charcoal shows no effect on intestinal gas and diarrhea and is ordinarily, medically ineffective against poisoning of strong acid or petroleum products and is particularly ineffective against poisonings of strong acids or bases, cyanide, iron, lithium, arsenic, methanol, ethanol or ethylene glycol. Activated carbon will not prevent these chemicals from being absorbed into the human body. It is on the World Health Organization's list of Essential Medicines.

ANALYTICAL CHEMISTRY

Activated carbon in 50% combination with celite, is used as stationary phase in low-pressure chromatographic separation of carbohydrates using ethanol solutions as mobile phase in analytical or preparative protocols. Activated carbon is useful for extracting the direct oral anticoagulants such as dabigatran, apixaban, rivaroxban and edoxabon from blood plasma samples. For this purpose it has been made into "minitablets" each containing 5mg activated carbon for treating 1ml samples of DOAC. Since this activated carbon has no effect on blood clotting factors, heparin or most other anticoagulants this allows a plasma sample to be analysed for abnormalities affected by the DOACs.

GAS PURIFICATION

Filters with activated carbon are usually used in compressed air and gas purification to remove oil vapors, odor, and other hydrocarbons from the air. The most common designs use a 1 stage or 2 stage filtration principle in which activated carbon is embedded inside the filter media. Activated carbon filters are used to retain radioactive gases within the air vacummed from a nuclear boiling water reactor turbine condenser. The large charcoal beds adsorb these gases and retain them while they rapidly decay to non-radioactive solid species. The solids are trapped in the charcoal practices, while the filtered air passes through.

FOOD ADDITIVE

Activated food grade charcoal became a food trend in 2016 being used as an additive to impart a slightly smoky taste and a dark coloring to products including hotdogs, ice cream, pizza bases and bagels. People taking medication including birth control pills and antidepressants, are advised to avoid novelty foods or drinks that use activated charcoal coloring, as it can render the medication ineffective.

STRUCTURE OF ACTIVATED CARBON

The structured of activated carbon has long been a subject of debate. In a book published in 2006. Harry Marsh and Francisco Rodriguez Reinoso considered more than 15 models for the structure, without coming to a definite conclusion about which was correct. Recent work using aberration-corrected transmission electron microscopy has suggested that activated carbons may have a structure related to that of the fullerenes, with pentagonal and heptagonal carbon rings.

2.2 INDUSTRY PROFILE

Coconut shell has a high quality natural structure, which is easy to form a well-developed microporous structure, thus forming a huge effective adsorption surface area, and research shows that each gram of coconut shell activated carbon has a surface area of about 500-1500 square meters. The activated carbon made from coconut shell is rich in high fiber, high density, uniform distribution of capillary pores and strong adsorption, which is most suitable for purifying air and drinking water. Besides, coconut shell activated carbon can also be used to extract gasoline from natural gas, purify kerosene, acetylene and kerosene, filter toxic gases, and make dry batteries and cigarette filters. In addition, coconut shell activated carbon also be used for decolorization and refinement, and purification in sugar chemical, food, reagent, and environmental protection industries.

On the one hand, environmental pollution has become a global problem, and activated carbon is increasingly used in air purification and water purification; on the other hand, although economic development has entered a new track and economic growth has slowed down, the overall development of industrial scale is still expanding, and the demand for activated carbon is also expanding.

The global activated carbon market size was valued at USD 3.44 billion in 2023 and is anticipated to expand at a compound annual growth rate (CAGR) of 2.9% from 2024 to 2032. The main growth driver is predicted to be increased demand for water treatment and sewage treatment applications. The presence of a significant number of small pores on the surface of activated carbon, also known as activated charcoal, distinguishes it. The pores increase the surface area of the element, which increases its adsorption capacity. It is frequently utilized in potable water filtration and sewage treatment plants, among other applications.

Manufacturing technical developments are projected to boost growth even more. The key raw material used for manufacturing the product is a coconut shell, resin, wood, and viscose rayon. When a huge quantity of raw material is used in the manufacturing process, the carbon yield is high.

For the end use industry, the water treatment industry dominated other industries, with a revenue share of 42.82%. This can be attributed to activated carbon's capacity to remove particulate and dissolved pollutants. It also removes contaminants and chlorine from water. It is used to purify the air to remove hazardous gases, smells, and harmful dust particles in greenhouses and manufacturing plants. The product can also be used in the food and beverages processing, Pharmaceutical and medical, and automotive industries.

CHAPTER-3

3.1 COMPANY PROFILE

Incorporated in 2010 and commencing operations in October 2011, Nova Carbons India Private Limited is an export oriented unit under Madras export promotion zone and is engaged in the manufacture of coconut shell based steam activated carbon. The company's shares were held by a private equity firm Constellation Holdings GmbH (Constellation), which is a 100.0% subsidiary of Jacobi Carbons AG and its managing director Mr.A.Antony Thomas and his wife. Jacobi Carbons AG belongs to the Jacobi Groups which is the largest manufacturer of coconut shell based activated carbon. In 2013-14, Japan based Osaka Gas Chemicals company limited acquired the Jacobi Groups, subsequent to which the shares of Mr.A.Antony Thomas and his wife were transferred to Osaka. The company's manufacturing facility is located in Tirunelveli.

Name of the organization : Nova Carbons India Pvt.Ltd

Date of Incorporation : 30th July 2010

Location : SIPCOT Industry, Gangaikondan, Tirunelveli.

Nature of Business : Manufacturing, Exporting

Type of Business : B2B

Product : Activated Carbon

Capacity : 1250 metric tons (MT) per month

Managing Director : A. Antony Thomas

Employees : 630 members

MISSION :

Be perceived as the preferred and most sustainable partner for solutions in purification process using Activated Carbon (AC), Ion exchange refiner and filter services.

VISION :

Be the number one Activated Carbon

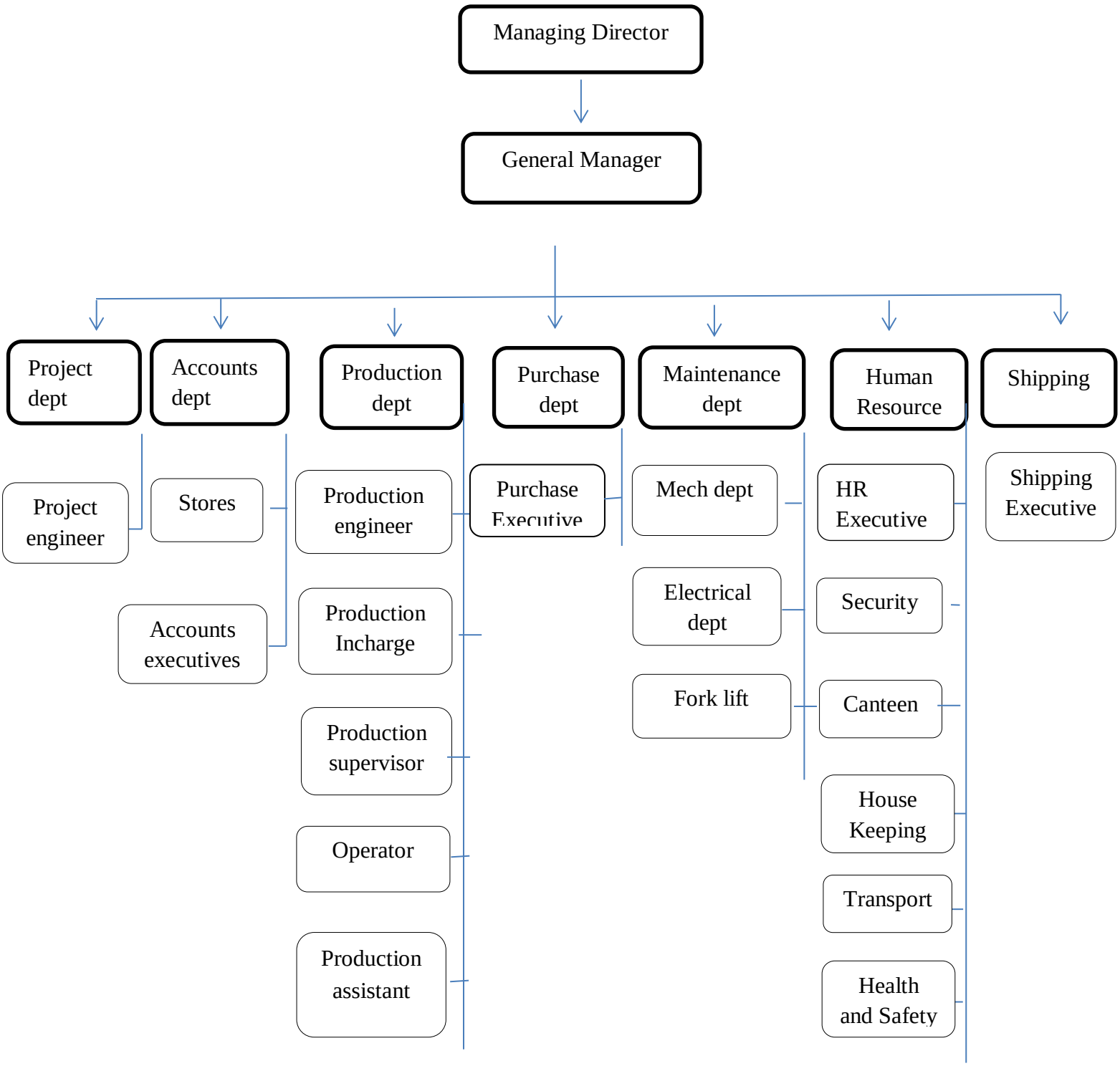
STRATEGY :

The path to make our vision come true, Living our mission, market focus, customer focus, expertise, service, acquisitions, partnership and people.

POLICIES AND PROCEDURES :

- Accounting policies : All accounting policies is followed by Indian Accounting Standard. As per the direction of Government Department, the organization maintain the books of accounts.
- Employee Benefits : As per the Government rules, the organization is giving the facilities of EPF, ESIC, Leaves etc.
- Quality Policy : Monitoring the effective realization of quality objectives and periodical review of its sustainability.
- Ensure effective and economical use of resources in the areas of energy, fuel, chemicals, water etc.
- Maintain absolute integrity at all times.
- Maintain dignity and discipline at all times.
- Treat people with respect and courtesy.
- Work in teams with a shared purpose and value individual ability and diversity.
- The organization also maintain all rules and regulations which is set by Indian Government.

3.2 ORGANIZATIONAL STRUCTURE



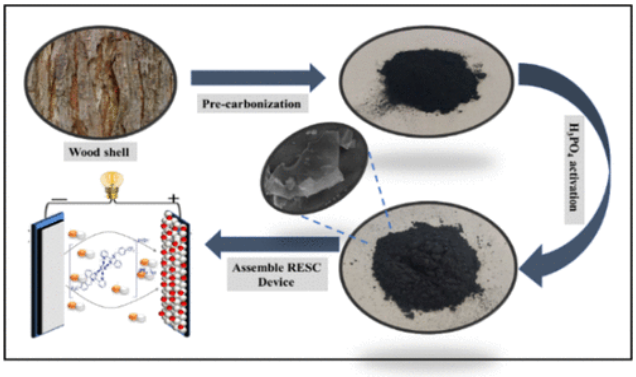
3.3 PRODUCT PROFILE

ACTIVATED CARBON FROM COCONUT SHELLS

Coconut shells can be used to produce high performance activated carbon (AC) electrodes for energy storage super capacitors. An incentive to promote this manufacturing route is sought through its anticipated positive impact on the environment and the substitution of non- renewable resources. The present study sets out to assess the environmental performance of a recently developed route for AC manufacturing through a simple activation process using steam. The analysis was carried out using the life cycle assessment (LCA) approach to investigate the production of AC material and resulting electrodes for a broad range of environmental impact categories and energy use. The study was conducted for a hypothetical optimized industrial-scale scenario drawing on experimental observations, literature, and energy and material balance calculations. Impact assessment results were presented both for the functional unit of electrode's capacitance and mass of AC and electrode, and interpreted through comparative analyses with coal-derived AC, reduced graphene oxide, and algae-derived biochar aerogel electrodes. The impact assessment results of the new AC electrode show competitive performance across most of the investigated impact categories and indicators. Larger impacts are mostly only observed for the land and water use categories stemming from the agriculturally intensive practice of coconut production. A total of 5.68 kg of CO_2 and 34.4 MJ of CED kg^{-1} AC are reported, with AC constituting roughly 60% of the total impacts arising in the production of super capacitor electrodes. The results have to be interpreted with present limitations to data especially considering the potentially high variability of carbon content in coconut shell species.

ACTIVATED CARBON FROM WOOD

The search for new superior electrode materials and modifications of electrolyte portions is one of the most significant responsibilities in developing super capacitors. To enhance the cell voltage and efficiency of super capacitors, we propose using activated Samanea saman wood shell carbon as an electrode and a new unique copper metal complex as an electrolyte for the first time. We synthesized a sheet-like structure of activated Samanea saman wood shell carbon using a pyrolysis process and synthesized a new novel copper metal complex bound with the H_2SO_4 electrolyte in this combination fabrication for a super capacitor (RESC). The redox couple aided SC exhibits an outstanding electrochemical performance that is attributed to the maximum specific capacitance value of 599 F g^{-1} and long lifespan of 98.9% specific capacitance after 8500 charge–discharge cycles in the half (three-electrode)-cell configuration. The activated carbon as cathode, graphite as anode, and gel formation of the copper metal complex bound with H_2SO_4 electrolyte were used to construct the (RESC) super capacitor device. This RESC device achieved 70.7 W h kg^{-1} energy density and power density of 534.75 W kg^{-1}. The combination of activated carbon electrode material and copper metal complex mediator provided the way for the development of a more efficient material for super capacitors (SCs).



APPLICATIONS OF ACTIVATED CARBON

- Metals Recovery

  Activated carbon is a valuable tool in the recovery of precious metals such as gold and silver.

- Food and Beverages

  Activated carbon is widely used throughout the food and beverage industry to accomplish a number of objectives.

- Medical
- Air Emission Purifications
- Biogas Purification
- Waste water (purification with activated carbon). etc.,

CHAPTER-4

FUNCTIONAL DEPARTMENTS

The organization maintaining various departments

4.1 PROJECT DEPARTMENT

- Conducting site investigations and feasibility studies.
- Preparing and managing the project budget.
- Developing and managing the project scope and timelines.
- Producing progress reports on a daily, weekly, and monthly basis.
- Requesting and selecting proposals from sub-contractors, including contract negotiations.
- Identifying potential project risks and solutions.
- Managing relationships with internal and external stakeholders.
- Creating a handover document for the operations and maintenance team.

4.2 ACCOUNTS DEPARTMENT

- Money out – Making payments and keeping the bills paid.
- Money in – Processing incoming payments.
- Payroll – Make sure everyone gets paid
- Reporting – Preparing financial reports, e.g P&L, Balance sheet and budget.
- Financial controls – To avoid errors, frauds and theft.

4.3 PRODUCTION DEPARTMENT

- **PULVERIZER DEPARTMENT**

Raw material to be pulverized enters the crushing chamber through the hopper and automatic rotary feeder. The impact of the hammer on the feed material against the liner plates reduces in into fine powder.

- **ETP (EFFLUENT TREATMENT PLANTS)**

Effluent Treatment plants or (ETP) are used by most of the companies in various industries to clean water and remove toxic and non-toxic materials or chemicals from it so that water can be reused or released in the environment which will do less harm to the environment.

- **GRANULATION DEPARTMENT**

Granular activated carbon is made from raw organic materials, such as coconut shells or coal, Which are high in carbon. Heat is used to activated to the surface area of the carbon. The activated carbon removes certain chemicals from the water passing through a GAC filter by trapping the chemical in GAC.

- **T-REX (TRASH EXTRACTION)**

It improves fine cleaning by using additional extraction points in the operating area of the cylinder. With great reliability it removes trash, dust and short fibers.

- **WASHING DEPARTMENT**

Activated carbons washed which hot water to remove excess phosphorus components until wash water attain PH of 7 followed by drying it.

- **PACKAGING DEPARTMENT**

Activated carbon packets use activated carbon to control objectionable odors and gases through the process of adsorption.

- **QUALITY CHECKING DEPARTMENT**

The properties exhibited by activated carbon include high surface area, variable pore size and volume, chemical intertness and stability.

- **HEALTH & SAFETY DEPARTMENT**

According to the Act, all factories must be kept clean, and all essential safeguards must be taken to safeguard the health of workers. Safety officers conduct routine workplace safety inspections to identify potential hazards, along with observing workers to ensure they're following safety procedures while on the job.

- **STORES**

Store keeping's basic functions is to receive the materials, recognize, place the same and issue the raw materials on the requisitions made by the respective departments.

4.4 FINANCE DEPARTMENT

SOURCES OF FINANCE

Finance is one of the major pillars of any organization and an essential ingredient to a successful business. Nowadays, a finance department has a broad range of roles to carry out within or outside an organization. The performance and success of any company greatly depend how well the finance is handled. Keeping a close watch on the financing function is very important for the smooth operation of a company. In this write-up, therefore, I will be reviewing the roles and responsibility of "finance" and "finance department" have Finance Department is the part of an organization that is responsible for acquiring funds for the firm, managing funds within the organization and planning for the expenditure of funds on various asset. It is the part of an organization that ensures efficient financial management and financial control necessary to support all business activities.

The main work of this department is to record the financial transactions. The day- to-day transactions are recorded in the proper book and are kept for reference. The transaction are first recorded in the journal as they take place.

A finance department manages a firm's long-term and day-to-day monetary operations and strategy. Finance groups oversee incoming and outgoing payments, budget creations, cash management (treasury), accounting, financial reporting and many other tasks related to the finances of the company.

FINANCE MANAGER FUNCTIONS

A financial manager is a person who takes care of all the important financial functions of an organization. The person in charge should maintain a far sightedness in order to ensure that the funds are utilized in the most efficient manner. His actions directly affect the profitability, growth and goodwill of the firm.

RAISING OF FUNDS

In order to meet the obligations of the business it is important to have enough cash and liquidity A firm can raise funds by the way of equity and debt. It is the responsibility of a financial manager to decide the ratio between debt and equity. It is important to maintain a good balance between equity and debt.

ALLOCATION OF FUNDS

Once the funds are raised through different channels the next important function is allocate the funds. The funds should be allocated in such a manner that they are optimally used. In order to allocate funds in the best possible manner the following point must be considered.

The size of the firm and its growth capability status of assets whether they are long-term or short-term mode by which the funds are raised these financial decisions directly and indirectly influence other managerial activities. Hence formation of a good asset mix and proper allocation of funds is one of the most important activity.

4.5 HUMAN RESOURCE DEPARTMENT

Human resource department handles a range of different functions within an organization. The department is responsible for hiring and firing employees, training workers, maintain interoffice relationships and interpreting employment laws. The department works diligently behind the scenes to ensure an organization runs efficiently.

NUMBER OF EMPLOYEES

The total number of employees in company is 630 employees. Male workers are 590 and female workers are 40. Office working hours from 8.30 am to 5.30 pm.

RECRUITMENT PROCESS

Recruitment refers to the process of identifying and attracting job seekers so as to build a pool of qualified job applicants. The process for interrelated stages,

1. Planning
2. Strategy development
3. Searching
4. Screening
5. Evaluation and control

TRAINING

In handling both employee needs and business needs in terms of having our workforce respond correctly, the company undertakes to conduct activity, general and technical training. The company have conducted internal training programs as well as the external training programs for all staff and worksite employees on various subjects such as Housekeeping methods, safety. job safety analysis, Management system implementation, Quality control circle process etc., on a regular basis.

4.6 SHIPPING DEPARTMENTS

For export 500kg jumbo bags (super sacks) are Palletized and shrink wrapped. It can also be palletized and shrink wrapped for exports.

CHAPTER-5

5.1 SWOT ANALYSIS

A SWOT analysis is a technique used to determine and define your Strength, Weakness, Opportunities and Threats.

SWOT analysis can be applied to an entire company or organization, or individual projects within a single department. Most commonly, SWOT analysis are used at the organizational level to determine how closely a business is aligned with its growth trajectories and success benchmarks, but they can also be used to ascertain how well a particular project such as an online advertising compaign is performing according to initial projections.

Whatever you choose to call them, SWOT analyses are often presented as a grid-like matrix with four distinct quadrants – one representing each individual element. This presentation offers several benefits, such as identifying which elements are internal versus external, and displaying a wide range of data in an easy-to-read, predominantly visual format.

Factors affecting your business . Using a SWOT , you can

- Understanding your business better.
- Address weakness.
- Capitalize on opportunities.
- Take advantage of your strength.
- Develop business goals and strength for achieving them.

SWOT ANALYSIS FOR NOVA CARBONS INDIA PRIVATE LIMITED

5.1 STRENGTH

- It’s an international Japanese brand with manufacturing worldwide, they have a very strong product portfolio.
- They have strategic agreements and have taken good business initiatives to expend their business.
- They have strong R&D facility which has achieved a number of milestones in using natural and renewable sources for raw materials.

5.2 WAEKNESS

- Climate change is a single impact category.
- Market share not growing with more stiff competition.

5.3 OPPORTUNITIES

- They have growth opportunity in emerging economics like India, China, Brazil etc.
- Their successful branding as the most Eco-friendly carbon company may take them a long way.

5.4 THREATS

- Fluctuation of exchanging rates.
- Japanese company is not growing and has reached a maturity phase.
- There is stiff competition from national and international brands.

LEARNING FROM THE STUDY

I got a very good experience in this NOVA Carbons India Private Ltd company during my internship time, and gave me an enthusiasm and a great experience. And now I have courage that one day, I will become a good entrepreneur. I developed many skills and techniques to directly applicable in career. This project provide develop attitudes conductive to effective interpersonal relationships and adjusting from college to full time employment in specialization upon graduation. I acquire more good work habits from employees. Having an experience of a good organization has one huge advantage, it makes us more professional and professionalism is the key to success. It have main emphasis on factors like

- Communication skills
- Professionalism
- Presentation abilities and confidence
- Handling the stressful situation

I have learned more knowledge and skills from this study, for each and every departments.

- In finance department finance is the blood of an organization. I have learned how to utilize the funds and properly current assets used from the finance departments. I have learned in finance department, how the funds are procured and utilized, financial aspects such as cost control, budgetary control and management accounting etc.
- In production department I have learned in my summer training that has watched the granulation process. Production department related aspects like production planning, production control and so on.
- In Human Resource department how they managing the manpower in an organization and how they effectively utilize their manpower skills in required departments. I have learned how the HR Executives are balance their work as well as their life. I have learned how they handling the grievances and how they sought out the employees problems without any delay.

CONCLUTION

The field survey has been done to know the overall process of the Nova carbons India private limited. The quality of the product is good. The capital will be provided directly from the managing director. The Nova carbons India private limited company performs the business activity in a effective manner. Each department specializes in a specific area by enabling as better meet the diverse needs of the industry. The visits to various departments were very interactive and added awareness about the organizations atmosphere. Face to face interaction helped to understand the vital role placed by each departments and their function in the organization.

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